

## Kunj Patel

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### PROFESSIONAL SUMMARY

Software and Systems Engineer with 2+ years of experience developing, restoring and integrating desktop, backend, web, geospatial, and sensor-connected applications. Experienced in C++/Qt, Python, FastAPI, Django, Angular, PostgreSQL/PostGIS, real-time networking and Linux-based systems. Built solutions across radar processing, GIS, robotics, AI/ML and product-development domains. Strong at understanding complex existing codebases, delivering end-to-end features and improving software reliability through automated testing, runtime diagnostics and configurable system design.

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### EDUCATION

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| Bachelor of Technology, Electrical and Computer Science Engineering                         | 2020 - 2024 |
| Institute of Infrastructure, Technology, Research & Management (IITRAM), Ahmedabad, Gujarat | 6.87/10.00  |

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### PROFESSIONAL EXPERIENCE

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| <b>System Engineer</b> , PentaShield Technologies, Vadodara, Gujarat | June 2024– |
| Present                                                              |            |

- Restored and extended legacy C++/Qt applications, recovering user-interface workflows, database connectivity, GIS visualisation and real-time runtime behaviour.
- Integrated interconnected software components using UDP/TCP communication, PostgreSQL and Linux-based multi-process environments.
- Diagnosed and repaired end-to-end data flows across simulation, interface, database and visualisation components.
- Developed Python and FastAPI backend services with Angular interfaces for real-time data processing, playback, diagnostics and simulation.
- Implemented and tested ASTERIX CAT048 and CAT062 data-ingestion, routing, replay and live UDP-processing workflows.
- Developed geospatial and simulation components using WGS84 calculations, target geometry, range modelling and probabilistic detection.
- Integrated ROS2, OpenCV, Open3D, LiDAR and RGB-D cameras for robotic perception, mapping, localisation and autonomous-navigation experiments.
- Improved software reliability through automated testing, configurable runtime behaviour, structured logging and repeatable Linux build and startup procedures.

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| <b>Intern</b> , Gujarat Infrastructure Development Board, Gandhinagar, Gujarat | December 2023–April 2024 |
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- Supported the implementation and validation of fair-price shop geospatial data on the PM Gati Shakti Gujarat portal.
- Coordinated a PM Gati Shakti seminar during Vibrant Gujarat for more than 350 participants and supported the launch of the Gujarat Compendium.

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| <b>Research Fellow</b> , IITRAM, Ahmedabad | May 2023–July 2023 |
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- Assembled, configured, and tested unmanned aerial vehicles using Mission Planner, QGroundControl, and ArduPilot.
  - Developed sensor-integration experiments using ultrasonic sensors and 3D LiDAR for obstacle detection, collision avoidance, and environmental perception.
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## SELECTED PROJECTS

### CarSetu — Used-Car Marketplace and Android Field-Capture Platform

Vite, Node.js, MongoDB Atlas, Kotlin, REST APIs, Vercel and Render 2026

- Developed a marketplace platform supporting admin-managed dealer onboarding, vehicle inventory, buyer leads and WhatsApp/call-based customer interactions.
- Built REST API workflows for dealer, vehicle, inventory, user and lead-management operations using MongoDB Atlas.
- Developed an Android field-capture application supporting vehicle creation and a structured 52-photo capture workflow.
- Implemented pending, approved and rejected photo statuses with an administrative review queue and rejection-reason handling.
- Deployed the frontend and backend using Vercel and Render with a cloud-hosted database.

### Radar Detection and Terrain-Aware Simulation Platform

Angular, Python, FastAPI, UDP, PostgreSQL, pytest and Geospatial Modelling 2026

- Developed a radar-processing and simulation platform supporting detections, tracks, scenarios, playback, diagnostics and live-data visualisation.
- Implemented ingestion, routing and replay workflows for ASTERIX CAT048 and CAT062 data, including live UDP processing.
- Developed terrain-aware simulation components covering WGS84 geometry, Earth curvature, horizon, slant range, signal-to-noise ratio and probabilistic detection.
- Added deterministic simulation behaviour and automated tests for physics components, configuration loading, runtime routing and end-to-end UDP workflows.

### TGBMC — GIS-Based Situational Awareness Platform

Angular, Luciad RIA, FastAPI, PostgreSQL, PostGIS, Redis and Docker 2026

- Architected the foundation of a GIS-based platform for geospatial visualisation, sensor awareness, operational planning, alerts and replay.
- Designed a PostgreSQL/PostGIS schema for geospatial data using WGS84 coordinates and SRID 4326.
- Structured the platform using Angular and Luciad RIA frontend components with FastAPI-based backend services.
- Configured Docker-based PostgreSQL/PostGIS infrastructure for repeatable local development and deployment.

### Autonomous Drone Object Detection and Geospatial Mapping

ROS, OpenCV, Gazebo, QGIS and Drone Simulation 2022–2023

- Developed a drone-based object-detection workflow under the IIT Bombay E-Yantra Sentinel Drone theme.
- Detected coloured objects from a simulated aerial camera while the drone followed a predefined flight path.
- Converted detected object positions into mapped geographic coordinates and visualised the results using QGIS.
- Secured 2nd position among more than 500 teams in the competition's hardware round.

## TECHNICAL SKILLS

- Programming Languages: C++, Python, TypeScript, JavaScript, SQL, Kotlin and MATLAB
- Frontend Development: Angular, React, Vite, Next.js, HTML5 and CSS3
- Backend Development: FastAPI, Django, Node.js, REST APIs and real-time backend services
- Desktop and GIS Development: Qt, Luciad RIA, QGIS and geospatial visualisation
- Databases and Data Systems: PostgreSQL, PostGIS, MongoDB Atlas and Redis

- AI, Computer Vision and Robotics: NumPy, pandas, scikit-learn, OpenCV, Open3D, TensorRT, ROS2 Humble, RViz2, Gazebo, SLAM, LiDAR and RGB-D processing
  - Networking and Systems Integration: UDP, TCP, JSON, API integration, PostgreSQL integration and Linux multi-process applications
  - Testing and Development Tools: pytest, API testing, regression testing, structured logging, Git, Linux, Docker, qmake, Make, npm and Gradle
  - Deployment and Infrastructure: Vercel, Render, Docker, environment-based configuration and cloud-hosted database integration
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## **ACHIEVEMENTS AND LEADERSHIP**

- Secured 2nd position among more than 500 teams in the hardware round of the IIT Bombay E-Yantra Robotics Competition.
- Led an IITRAM blood-donation drive that collected 310 units of blood.
- Coordinated trekking expeditions at PeakHikers India, managing participant safety, logistics and team coordination.